14.4.

⇒ Chain Rules with one / single variable:

→ The chain rule helps us find the derivative of a function inside another function. It is used when one function is nested inside another.

like. $y = f(g(x))$

Then to find dy/dx, we differentiate the outer function first, the multiply by the derivative of the inner function.

⇒ Chain Rule formula:

$$\frac{dy}{dx} = f'(g(x)) \times g'(x)$$

example (very simple)

$$y = (2x+3)^2$$

Here

- Outer function :- $f(u) = u^2$
- inner function :- $g(x) = (2x+3)$

Now apply chain Rule:

$$\frac{dy}{dx} = 2(2x+3) \times (2)$$

$$\Rightarrow \frac{dy}{dx} = 4(2x+3)$$

Ans.

ex2: (A bit more nested)

$$y = \sin(3x^2)$$

Here:

Outer: $f(u) = \sin(u)$

Inner: $g(x) = 3x^2$

Apply chain rule:

$$\frac{dy}{dx} = \cos(3x^2) \times 6x$$

$$\frac{dy}{dx} = 6x \cos(3x^2)$$